MATH 327 Homework 2 Fall 2026

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1. Problem 4.12

Problem. Let $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$ denote the set of irrational numbers. Prove that if $a, b \in \mathbb{R}$ with $a < b$, then there exists $x \in \mathbb{I}$ such that $a < x < b$. (Hint: first show that $r + \sqrt{2}$ is irrational for every $r \in \mathbb{Q}$.)

Lemma 1. *Let $r \in \mathbb{Q}$. Then $r + \sqrt{2} \notin \mathbb{Q}$.*

Proof. Suppose for the sake of contradiction that $r + \sqrt{2} \in \mathbb{Q}$. Thus, $r + \sqrt{2} = \frac{p}{q}$ for some $p \in \mathbb{Z}$, $q \in \mathbb{N}$, and $r = \frac{w}{v}$ for $w \in \mathbb{Z}$ and $v \in \mathbb{N}$. Hence, we have:

$$\begin{aligned} \frac{w}{v} + \sqrt{2} &= \frac{p}{q}, \\ \sqrt{2} &= \frac{p}{q} - \frac{w}{v}, \\ \sqrt{2} &= \frac{pv - wq}{qv}, \end{aligned}$$

where $(pv - wq), (qv) \in \mathbb{Z}$ under the additive and multiplicative closure of $\mathbb{Z}$.

Hence, a contradiction, and thus $r + \sqrt{2} \notin \mathbb{Q}$. □

Theorem 1. *If $a, b \in \mathbb{R}$ s.t. $a < b$, then there exists $x \in \mathbb{R} \setminus \mathbb{Q}$ s.t. $a < x < b$.*

Proof. Let $a, b \in \mathbb{R}$ s.t. $a < b$. Thus, we have:

$$\begin{aligned} a &< b, \\ 0 &< b - a, \\ 0 &< \frac{b-a}{2}, \\ a &< \frac{b-a}{2} + a \end{aligned} \tag{1}$$

and we have

$$\begin{aligned} a &< b, \\ a + b &< 2b, \\ b - a &< 2b - 2a, \\ \frac{b-a}{2} &< b - a, \\ \frac{b-a}{2} + a &< b. \end{aligned}$$

Thus, we have

$$a < \frac{b-a}{2} + a < b \tag{2}$$

via transitivity.

Since $\frac{b-a}{2} > 0$ by inequality (1), and since $\sqrt{2} > 0$, then by the Archimedean Property, $\exists\, n \in \mathbb{N}$ s.t. $n\left(\frac{b-a}{2}\right) > \sqrt{2}$,

$$\frac{b-a}{2} > \frac{\sqrt{2}}{n}. \tag{3}$$

Let $n \in \mathbb{N}$ be such n.

Since $\sqrt{2}, n > 0$, then $\frac{\sqrt{2}}{n} > 0$ and hence:

$$\frac{\sqrt{2}}{n} + a > a \tag{4}$$

Via transitivity of inequalities (2), (3), and (4) we have:

$$\begin{aligned} \frac{b-a}{2} + a &> \frac{\sqrt{2}}{n} + a > a, \\ b &> \frac{\sqrt{2}}{n} + a > a, \end{aligned}$$

and by Lemma 1, $\frac{\sqrt{2}}{n} + a \notin \mathbb{Q}$.

Hence, $\frac{\sqrt{2}}{n} + a \in \mathbb{I}$ s.t. $a < \frac{\sqrt{2}}{n} + a < b$. □

2. Problem 4.14(b)

Problem. Let $A, B \subseteq \mathbb{R}$ s.t. A and B are nonempty and bounded. Let $A + B$ be the set of all sums $a + b$ where $a \in A$ and $b \in B$. Prove $\inf(A + B) = \inf(A) + \inf(B)$.

Proof. Suppose that $\inf(A) + \inf(B) \neq \inf(A + B)$. Thus, $\inf(A) + \inf(B)$ is not a lower bound for $A + B$, or $\exists$ a lower bound for $A + B$ less than $\inf(A) + \inf(B)$.

Case 1. Suppose for the sake of contradiction that $\inf(A) + \inf(B)$ is not a lower bound for $A + B$. Thus, $\exists\, a + b \in A + B$ s.t. $a + b < \inf(A) + \inf(B)$. Since $a \geq \inf(A)\ \forall a \in A$, then $0 \geq \inf(A) - a$. Hence,

$$b < (\inf(A) - a) + \inf(B) \leq \inf(B),$$

where $b < \inf(B)$ contradicts the def. of an infimum, and thus, $\inf(A) + \inf(B)$ is a lower bound for $A + B$.

Case 2. Suppose for the sake of contradiction that $\inf(A) + \inf(B)$ is not the greatest lower bound for $A + B$. Thus, $\forall\, a + b \in A + B$,

$$a + b \geq \inf(A + B) > \inf(A) + \inf(B).$$

Since $\inf(A + B) > \inf(A) + \inf(B)$, then there exists $x \in \mathbb{R}$ s.t.

$$\inf(A) + \inf(B) < x < \inf(A + B) \leq a + b.$$

Thus, we have:

$$\begin{aligned} \inf(A) + \inf(B) < x < a + b, \\ \inf(A) + (\inf(B) - b) < x - b < a, \end{aligned}$$

where $\inf(B) \leq b$, or $\inf(B) - b \leq 0$.

Suppose $\inf(B) = b$, or $\inf(B) - b = 0$. Thus, we have:

$$\inf(A) + (\inf(B) - b) = \inf(A) < x - b < a,$$

where $x - b$ is a greater lower bound for A than $\inf(A)$, which is a contradiction.

Hence, we're left with $\inf(B) < b$. Thus, we have:

$$\inf(B) < \frac{b - \inf(B)}{2} + \inf(B) < b,$$

where $\frac{b - \inf(B)}{2} + \inf(B)$ is a greater lower bound for B than $\inf(B)$, and we've thus exhausted all options with contradictions.

Hence, $\inf(A + B) = \inf(A) + \inf(B)$. □

3. Problem 8.2(c)

Problem. Determine the limit of $c_n = \frac{4n+3}{7n-5}$ and prove that claim.

Theorem 2. $\lim \left[\frac{4n+3}{7n-5}\right] = \frac{4}{7}$.

Proof. Let $c_n = \frac{4n+3}{7n-5}$. Let $\varepsilon > 0$. Let $N \in \mathbb{N}$ s.t. $\frac{1+35\varepsilon}{49\varepsilon} < N$.

Thus, we have:

$$\begin{aligned}
\frac{1+35\varepsilon}{49\varepsilon} &< N, \\
\frac{1+35\varepsilon}{\varepsilon} &< 49N, \\
\frac{1}{\varepsilon} + 35 &< 49N, \\
\frac{1}{\varepsilon} &< 49N - 35, \\
\frac{1}{49N-35} &< \varepsilon,
\end{aligned}$$

and since $N \in \mathbb{N}$, then $49N - 35 \geq 0$. Hence, $\left|\frac{1}{49N-35}\right| = \frac{1}{49N-35}$.

Thus, we have:

$$\begin{aligned}
\left|\frac{1}{49N-35}\right| &< \varepsilon, \\
\left|\frac{(21-20)+(28N-28N)}{49N-35}\right| &< \varepsilon, \\
\left|\frac{(28N+21)-4(7N+5)}{49N-35}\right| &< \varepsilon, \\
\left|\frac{7(4N+3)-4(7N-5)}{7(7N-5)}\right| &< \varepsilon, \\
\left|\frac{4N+3}{7N-5} - \frac{4}{7}\right| &< \varepsilon.
\end{aligned}$$

Thus, for an arbitrary $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t. $n \geq N$ implies:

$$\left|\frac{4N+3}{7N-5} - \frac{4}{7}\right| < \varepsilon.$$

Hence, $\lim \left[\frac{4n+3}{7n-5}\right] = \frac{4}{7}$. □

4. Problem 8.4

Problem. Let (t_n) be a bounded sequence, i.e., there exists M s.t. $|t_n| \leq M$ for all n, and let (s_n) be a sequence s.t. $\lim s_n = 0$. Prove that $\lim(s_n t_n) = 0$.

Proof. Let s_n be a sequence s.t. $\lim[s_n] = 0$. Let (t_n) be a bounded sequence s.t. $|t_n| \leq M$ for all $n \in \mathbb{N}$. Suppose for the sake of contradiction that $s_n t_n$ diverges.

Thus, there exists $\varepsilon > 0$ s.t. $\forall N \in \mathbb{N}$, $n \geq N$ and $|s_n t_n| \geq \varepsilon$.

Let $\varepsilon_2 > 0$ be such that

$$|s_n| M \geq \varepsilon_2,$$
$$|s_n| \geq \frac{\varepsilon_2}{M}.$$

Let $\varepsilon_1 = \frac{\varepsilon_2}{M}$. Since $\varepsilon_2 > 0$ and $0 \leq |t_n| \leq M$, then $\frac{\varepsilon_2}{M} \geq 0$.

Thus, converging to 0 implies that $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ s.t. $n \geq N$ implies $|s_n| < \varepsilon$.

However, ε_2 is an $\varepsilon > 0$ s.t. $|s_n| \geq \varepsilon$. Hence, a contradiction. □

5. Problem 8.8(a)

Problem. Prove $\lim\left[\sqrt{n^2+1}-n\right]=0$.

Proof. Let $\varepsilon>0$. By the Archimedean Property, there exists $n\in\mathbb{N}$ s.t.

$$\frac{1-\varepsilon^2}{2\varepsilon}<n.$$

Let N be such n. Thus, we have:

$$\begin{aligned}
\frac{1-\varepsilon^2}{2\varepsilon}&<N,\\
\frac{1-\varepsilon^2}{\varepsilon}&<2N,\\
\frac{1}{\varepsilon}-\varepsilon&<2N,\\
\frac{1}{\varepsilon}&<2N+\varepsilon,\\
1&<2N\varepsilon+\varepsilon^2,\\
N^2+1&<N^2+2N\varepsilon+\varepsilon^2,\\
\sqrt{N^2+1}&<\varepsilon+N,
\end{aligned}$$

since $\varepsilon, N>0$ so $\sqrt{N^2+1}>0$ and hence $\sqrt{N^2+1}<-\varepsilon-N$ is a contradiction.

Thus, we have:

$$\begin{aligned}
\sqrt{N^2+1}&<\varepsilon+N,\\
\sqrt{N^2+1}-N&<\varepsilon,\\
\sqrt{N^2\left(1+\frac{1}{N^2}\right)}-N&<\varepsilon,\\
N\sqrt{1+\frac{1}{N^2}}-N&<\varepsilon,
\end{aligned}$$

where $\sqrt{1+\frac{1}{N^2}}>1\ \forall N\in\mathbb{N}$. Hence, $N\sqrt{1+\frac{1}{N^2}}>N$ and thus $\sqrt{N^2+1}-N>0$.

Therefore,

$$\sqrt{N^2+1}-N=\left|\sqrt{N^2+1}-N\right|<\varepsilon,$$

where $\forall\varepsilon>0$, we've shown there exists $N\in\mathbb{N}$ s.t. $n\geq N$ implies

$$\left|\sqrt{N^2+1}-N\right|<\varepsilon,\quad\text{or}\quad\left|\left(\sqrt{N^2+1}-N\right)-0\right|<\varepsilon.$$

Hence, $\lim\left[\sqrt{n^2+1}-n\right]=0$. □

6. Problem 6

Problem. Prove that if (s_n) converges, then so does $(|s_n|)$. Show by an example that the converse of this statement does not hold.

Proof. Suppose a_n converges to a. Let ε_1 be such an ε s.t. $\frac{\varepsilon_1}{2} > 0$. Hence, there exists $N \in \mathbb{N}$ s.t. $n \geq N$ implies:

$$\begin{aligned}
|a_n - a| &< \frac{\varepsilon_1}{2}, \\
2|a_n - a| &< \varepsilon_1, \\
|a_n - a| + |a_n - a| &< \varepsilon_1, \\
|a_n - a| &< \varepsilon_1 - |a_n - a|, \\
|a_n - a| &< \varepsilon_1 - \big|(a_n - |a_n|) + (|a_n| - a)\big| \\
&\leq \varepsilon_1 - \big|a_n - |a_n|\big| - \big||a_n| - a\big|, \qquad (1)
\end{aligned}$$

where inequality (1) holds via the triangle inequality. Thus, via transitivity, we have:

$$\begin{aligned}
|a_n - a| &< \varepsilon_1 - \big|a_n - |a_n|\big| - \big||a_n| - a\big|, \\
|a_n - a| + \big|a_n - |a_n|\big| + \big||a_n| - a\big| &< \varepsilon_1.
\end{aligned}$$

Since $|a_n - a|, \big|a_n - |a_n|\big| \geq 0$, then

$$\big||a_n| - a\big| \leq |a_n - a| + \big|a_n - |a_n|\big| + \big||a_n| - a\big| < \varepsilon_1,$$

and via transitivity, we have:

$$\big||a_n| - a\big| < \varepsilon_1.$$

Thus, for all $\varepsilon_1 > 0$, there exists $N \in \mathbb{N}$ s.t. $n \geq N$ implies $\big||a_n| - a\big| < \varepsilon_1$. Hence, $|a_n|$ converges and thus, a_n converging implies $|a_n|$ converges. □

7. Problem 8.7(b)

Problem. Prove $S_n = (-1)^n n$ diverges.

Proof. Let $n \in \mathbb{N}$ and let $S_n = (-1)^n n$.

Case 1. Suppose $n = 2k$ for some $k \in \mathbb{Z}$. Thus,

$$(-1)^{2k} \cdot n = \left[(-1)^k\right]^2 \cdot n,$$

where $\forall x \in \mathbb{R}$, $x^2 \geq 0$. Thus since $n \geq 1$, then $\left[(-1)^k\right]^2 \cdot n \geq 0$, and hence, $(-1)^n = 1$ and $(-1)^n \cdot n = n$. Hence, $|(-1)^n \cdot n| = 1 \cdot n = n$.

Case 2. Suppose $n = 2j + 1$ for some $j \in \mathbb{Z}$. Thus,

$$(-1)^n \cdot n = (-1)^{2j+1} \cdot n = (-1)^{2j} \cdot (-1) \cdot n,$$

where $(-1)^{2j} \cdot n$ for $j \in \mathbb{Z}$ is equal to n. Thus,

$$(-1)^n \cdot n = (-1)^{2j} \cdot (-1) \cdot n = (-1) \cdot n = -n.$$

Hence, $|(-1)^n \cdot n| = |-n| = n$ since $n \geq 1$.

Thus, $|S_n| = |(-1)^n n| = n$.

Since $|S_n| = n$ is a trivially divergent sequence, then by Problem 6 of this homework (HW 2), the contrapositive says:

"If $|a_n|$ diverges, then a_n diverges."

Hence, since $|S_n| = n$ diverges, then $S_n = (-1)^n n$ diverges. □

8. Problem 8.9(a)

Problem. Let (s_n) be a convergent sequence. Show that if $s_n \geq a$ for all but finitely many n, then $\lim s_n \geq a$. (Per the assignment, you may assume the inequality holds for all n.)